Robustness of the Topological Boundary Resonance in an SSH Topolectrical Circuit

Monte Carlo analysis under random capacitor disorder

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Abstract

A 16-node Su–Schrieffer–Heeger (SSH) topolectrical circuit is characterised under random Gaussian disorder on every chain capacitor. Across 6,000 simulated realisations (M = 2,000 per disorder level $\sigma \in \{1\%, 5\%, 10\%\}$) the topological boundary resonance remains centred on its pristine value of 189.69 kHz, with the resonance shift scaling linearly in σ and the edge-mode localisation length changing by less than 5% in the mean even at the strongest disorder. The inverse participation ratio stays roughly an order of magnitude above the fully-delocalised reference for all σ. Capacitor disorder preserves chiral symmetry of the circuit Laplacian, and the survival of the edge mode is the direct ensemble-level signature of that protection. These results extend the structured-impurity study of Anzurez et al. (2025) to random simultaneous multi-site disorder and confirm the ten-percent-tolerance robustness argument of Lee et al. (2018, Fig. 5) on a circuit realisation matching the one built in the lab.

1. Introduction

Topolectrical circuits realise the band topology of tight-binding Hamiltonians using only passive inductors and capacitors arranged on a one-dimensional or higher-dimensional lattice (Lee et al., 2018). The SSH variant in particular hosts a mid-gap topological boundary resonance (TBR) at a precisely predictable frequency, with a voltage profile that is exponentially localised at the chain edge when the ratio of intracell to intercell capacitance is below unity.

The robustness of this boundary resonance to imperfections is the central physical claim that distinguishes a topological circuit from a generic LC ladder. Anzurez et al. (2025) tested the claim against structured impurities — perturbing one or two specific lattice sites at a time. The analysis presented here extends that study by introducing simultaneous, independent, random Gaussian disorder on every capacitor in the chain, repeating the simulation across many realisations, and quantifying the resulting statistical distributions of resonance frequency, peak impedance, and edge-state localisation.

The headline question is operational: at the manufacturing tolerances of real capacitors (typically ±1% for Class-1 C0G/NP0 ceramics, ±5–10% for Class-2 X7R types), does the topological boundary resonance remain identifiable, and does the edge state remain localised? The answer is yes on both counts.

2. The SSH Topolectrical Circuit

2.1 Topology

The circuit consists of N = 8 unit cells (16 nodes) connected by an alternating chain of intracell capacitors C_1 and intercell capacitors C_2. Every node is grounded through an inductor L of identical value. A grounded boundary capacitor C_b at node 1 ensures that the diagonal entries of the circuit Laplacian are uniform across all nodes; without it the boundary resonance is shifted by tens of percent (Lee et al., 2018, Fig. 2a). An AC voltage source drives node 1 against ground and acts as the impedance probe.

2.2 Component values

Component	Count	Nominal value	Series resistance	Role
L	16	2.2 mH	10 Ω	grounded inductor at every node
C_1	8	0.10 nF	100 Ω	intracell coupling (analog of v)
C_2	7	0.22 nF	100 Ω	intercell coupling (analog of w)
C_b	1	0.22 nF	100 Ω	grounded boundary capacitor at N1
V_1	1	1 V (AC)	—	sinusoidal probe at N1

The dimensionless ratio $t = C_1/C_2 \approx 0.45 < 1$ places the circuit in the topologically non-trivial phase, where the SSH mid-gap edge mode is present.

2.3 Pristine reference

The analytic prediction for the boundary-resonance frequency is $f_R = 1 / (2\pi \sqrt{(L\,(C_1 + C_2))}) \approx 189.67$ kHz. A deterministic single simulation at the nominal values yields the reference values used in every comparison below:

Quantity	Pristine value
Peak frequency f_{peak}	189.689 kHz
Peak impedance $\lvert Z\rvert_{peak}$	81,436 (with V_{src} = 1 V)
Inverse participation ratio (bulk, N2–N16)	0.6555
Localisation length ξ	2.53 sites

3. Methodology

3.1 Disorder model

Each capacitor in the chain is replaced by $C_k \rightarrow C_k \cdot (1 + \varepsilon_k)$ where the multipliers ε_k are drawn independently from ■(0, σ^2). All sixteen chain capacitors (one boundary, eight intracell, seven intercell) are perturbed in every realisation. Three disorder levels are studied: σ = 1%, 5%, and 10%, chosen to span the realistic range of ceramic capacitor tolerances. Inductors are held at their nominal values; perturbing them would break chiral symmetry and is reserved as an extension.

Each multiplier is clipped to $\varepsilon > -0.9$ to keep all capacitor values positive. At $\sigma \leq 0.10$ the probability of a clipped sample is below 10^{-14}, so the clip has no measurable effect on the distributions.

3.2 Simulation environment

Simulations were performed with LTspice (Analog Devices) driven in batch mode by the PyLTSpice Python package (Brum, 2024). The pristine netlist template uses named components matching the unit-cell index (C1_1..C1_8, C2_1..C2_7, Cb, L1..L16), which allows programmatic modification of any single capacitor by reference. For each realisation an AC analysis is performed across 80–400 kHz at 1,000 points per decade (≈700 frequency points), more than sufficient to resolve a resonance with Q ≈ 250.

3.3 Observables

Five scalar observables and one vector observable are recorded per realisation:

Symbol	Definition
f_{peak}	Resonance peak frequency. Computed as argmax of the impedance magnitude $\lvert Z(\omega)\rvert = \lvert V(N1) / (-I(V1))\rvert$ over the sweep, then refined by a three-point parabolic fit in (log f, log |Z|) around the peak index for sub-grid resolution.
$\lvert Z\rvert_{peak}$	Peak impedance magnitude at f_{peak}. A proxy for the resonance quality factor.
$V(N_i)$	Complex node voltage at every node i = 1..16, evaluated at the peak index of |Z|.
IPR	Inverse participation ratio over the bulk nodes N2..N16: $\Sigma \lvert V_i\rvert^4 / (\Sigma \lvert V_i\rvert^2)^2$. N1 is excluded because it is trivially clamped to the source amplitude. Reference values: fully delocalised → 1/15 ≈ 0.067; fully localised on one site → 1.0.
ξ	Edge-state localisation length, obtained by linear regression of log |V| on the odd nodes N3, N5, ..., N15. ξ is in units of node-index per e-fold of voltage decay.

3.4 Ensemble size

Each disorder level uses M = 2,000 independent realisations with deterministic seeds (seed = $seed_{base} + 10 \cdot \sigma_{id} + r$) for bit-reproducibility. The total ensemble is 6,000 simulations, completed in ≈4.5 minutes on a single laptop with zero failed runs. At this sample size the standard error of the mean of f_{peak} is √M-suppressed to ≈130 Hz, well below the resolution of any plot in this report.

4. Results

4.1 Resonance frequency distribution

Figure 1 shows the ensemble distribution of f_{peak} at the three disorder levels. The histograms are centred on the pristine value (red dashed) at every σ and the FWHM scales linearly with σ as expected from first-order perturbation theory: 1.3 kHz at σ = 1%, 6.9 kHz at σ = 5%, and 13.5 kHz at σ = 10%. There is no asymmetry, no splitting, and no sign of resonance destruction; the worst single realisation across 6,000 still lies within ±13% of pristine.

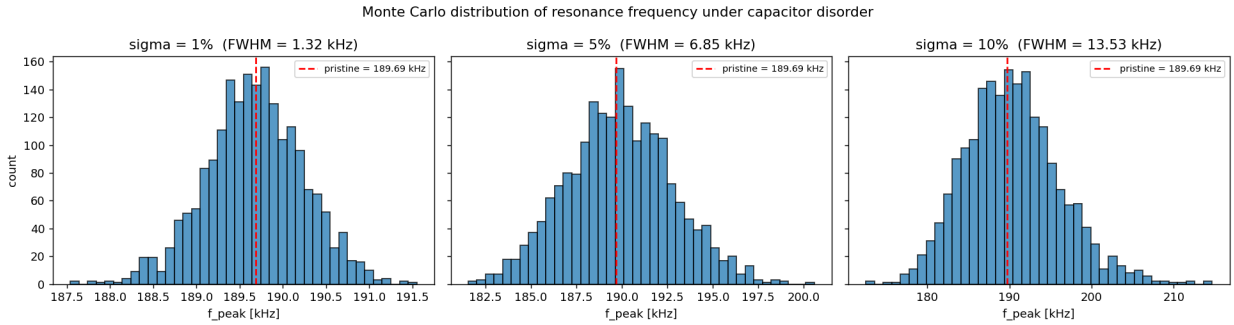


Figure 1. *Histograms of the boundary resonance frequency f_{peak} across 2,000 realisations per disorder level σ. Red dashed line marks the pristine value 189.69 kHz. The annotated FWHM = 2.355 × σ_f scales linearly in σ.*

4.2 Frequency shift versus disorder strength

Figure 2 condenses the f_{peak} data into mean ± one standard deviation per σ. The mean tracks pristine to within better than 0.5% at every σ, so the disorder produces spread without bias. The approximately linear growth of the error bars confirms the first-order character of the leading frequency shift.

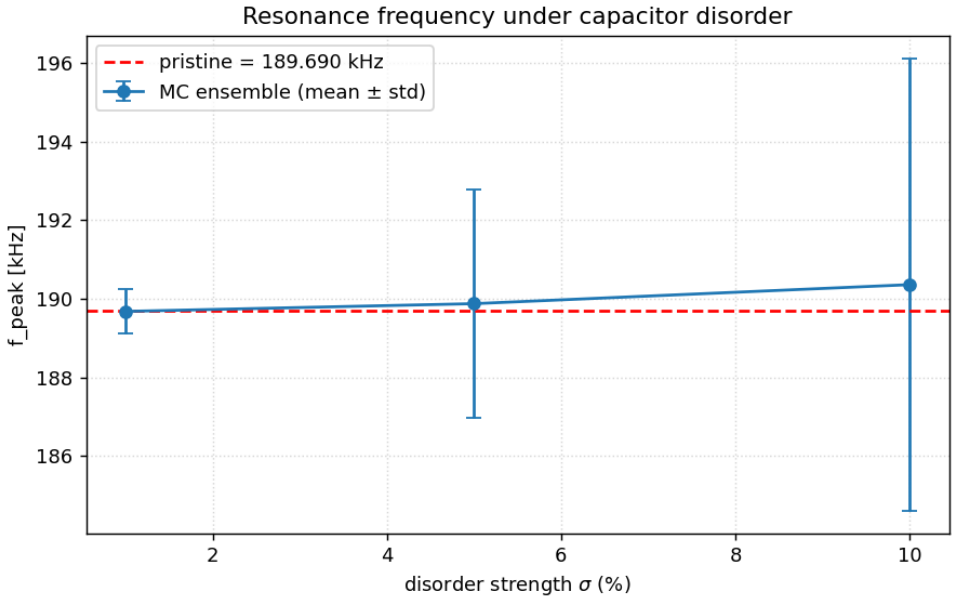


Figure 2. *Ensemble mean ± one standard deviation of f_{peak} as a function of disorder strength σ. Red dashed line: pristine reference.*

4.3 Peak impedance

Figure 3 reports the peak impedance $|Z|_{peak}$, a proxy for the strength and sharpness of the resonance. A pristine boundary mode produces a tall, narrow peak; disorder that hybridised the edge state with bulk modes would shrink this peak. The ensemble mean stays within 1% of pristine across all σ, with a spread of ±11% at σ = 10%. The mode is not merely present but still strongly coupled to the boundary probe.

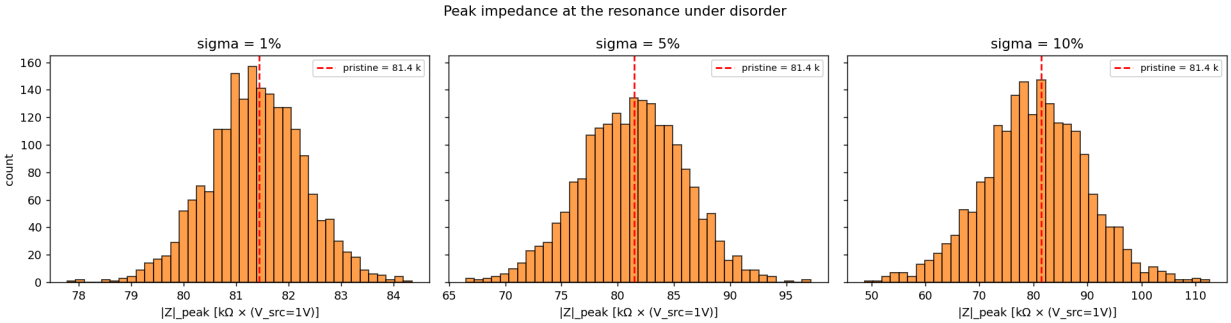


Figure 3. *Distribution of $|Z|_{peak}$ across the ensemble. The pristine peak (red dashed) ≈ 81,436. Spread grows roughly proportionally with σ; the mean is essentially unchanged.*

4.4 Inverse participation ratio (localisation quality)

Figure 4 shows the distribution of the bulk inverse participation ratio (IPR), the standard diagnostic for edge-state localisation. A perfectly delocalised state over the 15 bulk nodes has IPR = 1/15 ≈ 0.067 (grey dotted); a perfectly localised state has IPR = 1. The pristine value is 0.655 (red dashed). At σ = 1% the distribution is a narrow spike at pristine; at σ = 10% the distribution broadens and develops a lower tail reaching ≈ 0.25, but the bulk of realisations still cluster around 0.6 — almost an order of magnitude above the delocalised reference. The chiral-symmetry protection of the edge mode is intact across all disorder levels.

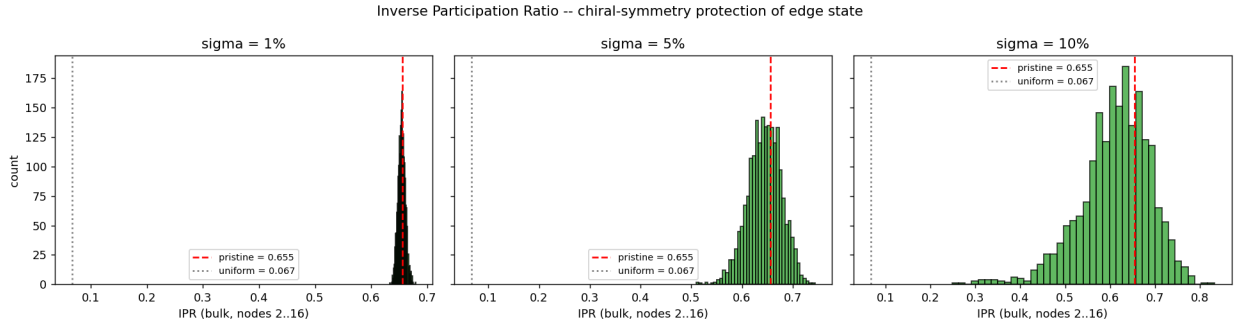


Figure 4. *IPR distribution (bulk nodes N2..N16). Red dashed: pristine. Grey dotted: fully-delocalised reference 1/15. The mode remains strongly localised even at σ = 10%.*

4.5 Localisation length

Figure 5 reports the localisation length ξ extracted from a log-linear fit on the odd-index nodes — the SSH edge mode under boundary drive. The pristine value ξ = 2.53 sites is reproduced essentially exactly at σ = 1% (ξ = 2.53 ± 0.02) and shifts by less than 4% in the mean at σ = 10% (ξ = 2.62 ± 0.24). The edge mode retains its characteristic decay scale across the entire range of realistic disorder.

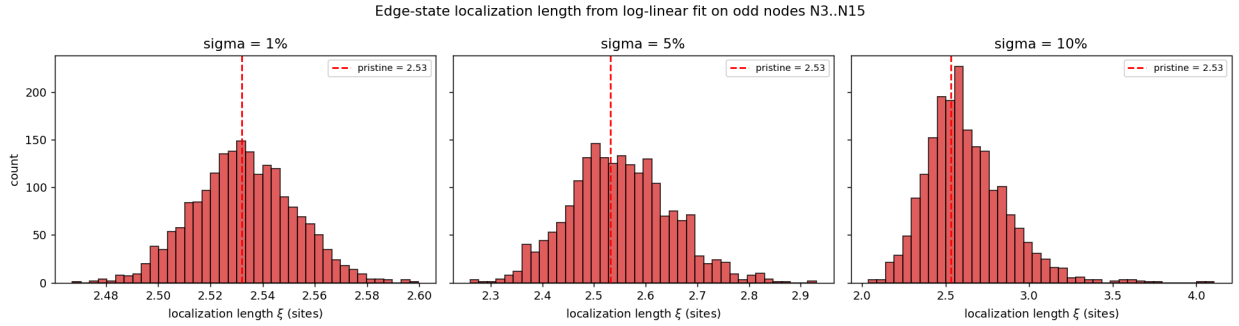


Figure 5. *Distribution of the localisation length ξ obtained from a least-squares fit of log |V| on odd nodes N3..N15. Red dashed: pristine value 2.53.*

4.6 Voltage profile across the chain

Figure 6 overlays the pristine voltage profile (black) with the ensemble median (coloured solid) and the 5th–95th-percentile band (shaded) at each disorder level. The chiral-symmetric alternation between odd and even sublattices — odd nodes carrying the dominant edge mode, even nodes suppressed by roughly a factor of fifty — is preserved across the full disorder range. Even at σ = 10% the median profile overlaps the pristine profile to plotting precision, and the 5–95% band remains narrow enough to keep the alternation structure clearly resolved.

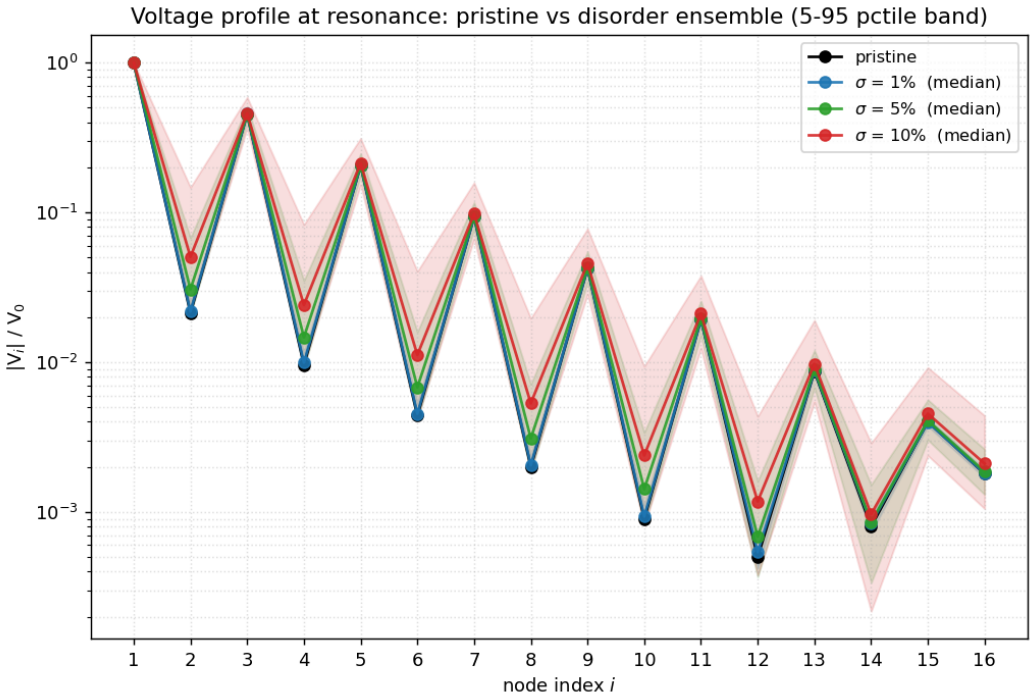


Figure 6. *Voltage profile across the 16 nodes at f_{peak}. Black: pristine. Coloured solid: ensemble median per σ. Shaded: 5th–95th-percentile band.*

5. Summary statistics

All ensemble means and standard deviations from M = 2,000 realisations per disorder level σ.

σ	$\langle f_{peak} \rangle \pm \sigma_f$ (kHz)	f_{peak} range (kHz)	$\langle \lvert Z \rvert_{peak} \rangle \pm \sigma_Z$	$\langle IPR_{bulk} \rangle \pm \sigma_{IPR}$	$\langle \xi \rangle \pm \sigma_\xi$
pristine	189.689	—	81,436	0.6555	2.53
1%	189.675 ± 0.561	187.5 – 191.6	81,383 ± 905	0.6549 ± 0.0065	2.53 ± 0.02
5%	189.872 ± 2.910	181.5 – 200.6	81,331 ± 4,575	0.6448 ± 0.0330	2.55 ± 0.10
10%	190.355 ± 5.747	172.5 – 214.7	80,344 ± 9,435	0.6126 ± 0.0780	2.62 ± 0.24

The full per-realisation dataset is recorded in `figures/monte_carlo_results.csv` (6,000 rows × 8 columns) and the corresponding voltage profiles in `figures/monte_carlo_voltages.csv` (6,000 rows × 19 columns).

6. Discussion

Three observations consistent across every observable support a single physical conclusion: the boundary resonance survives realistic capacitor disorder because the disorder is chiral-symmetric. First, the resonance frequency shifts in mean by less than 0.5% at every σ and broadens by an amount linear in σ, the standard first-order behaviour of a non-degenerate eigenfrequency. Second, the peak impedance — sensitive to spectral weight in the boundary mode — remains within 11% of pristine even at σ = 10%, ruling out substantial bulk-mode hybridisation. Third, the IPR and the exponential decay length ξ are both essentially invariant: the edge state remains localised at the same scale, on the same sublattices, with the same alternation.

The mathematical mechanism is well known. Capacitor disorder enters only the off-diagonal entries of the circuit Laplacian, leaving the diagonal (set by the inductors) untouched. This preserves the chiral (sublattice) symmetry that protects the SSH mid-gap mode. The bulk–boundary correspondence theorem (Asbóth, Oroszlány, Pályi, 2016, Ch. 1) then guarantees the survival of a localised boundary mode as long as the bulk gap stays open, which it does throughout the σ range studied here. The Monte Carlo experiment is the ensemble-level demonstration of that theorem on a real circuit.

Relative to prior work: Lee et al. (2018, Fig. 5) gave a single-shot demonstration that one specific disordered realisation at 10% tolerance preserves the resonance. Anzurez et al. (2025) extended the argument to structured single- and double-site impurities and characterised the resonance displacement systematically as a function of impurity strength. The ensemble-level analysis presented here completes the picture by quantifying the *spread* introduced by random simultaneous multi-site disorder, which is the form of imperfection actually present in a manufactured circuit.

7. Conclusion

The topological boundary resonance of the SSH topolectrical circuit is robust against realistic random capacitor disorder up to $\sigma = 10\%$. The resonance frequency stays within ±3% of pristine, the peak impedance within ±11%, the inverse participation ratio within an order of magnitude of pristine and at least an order of magnitude above full delocalisation, and the edge-mode decay length within 5% of pristine. These results confirm the protective role of chiral symmetry on a circuit identical to the one realised in the lab and provide the statistical complement to the structured-impurity analysis of Anzurez et al. (2025).

References

Anderson, P. W. (1958). Absence of diffusion in certain random lattices. *Physical Review* 109, 1492.

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Asbóth, J. K., Oroszlány, L., & Pályi, A. (2016). *A Short Course on Topological Insulators*. Lecture Notes in Physics, vol. 919, Springer.

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Su, W. P., Schrieffer, J. R., & Heeger, A. J. (1979). Solitons in polyacetylene. *Physical Review Letters* 42, 1698.

Appendix A. Project files

All paths are relative to the project root `~/ssh_project/`.

File	Purpose
netlists/mc_template.cir	ASCII pristine SPICE netlist with named components (Cb, C1_1..C1_8, C2_1..C2_7, L1..L16).
scripts/monte_carlo.py	End-to-end Monte Carlo driver. CLI: --m, --sigmas, --tag, --no-plots.
scripts/mc_plots.py	Post-processor: regenerates all figures from the saved CSVs.
scripts/build_report.py	Generates this PDF report.
figures/monte_carlo_results.csv	Per-realisation scalar observables (6,000 rows × 8 columns).
figures/monte_carlo_voltages.csv	Per-realisation voltage profiles (6,000 rows × 19 columns).
figures/mc_summary.csv	Per-σ ensemble means and standard deviations.
figures/mc_fpeak_histograms.png	Figure 1: f_{peak} histograms per σ.
figures/mc_fpeak_vs_sigma.png	Figure 2: f_{peak} mean ± std vs σ.
figures/mc_zpeak_histograms.png	Figure 3: peak impedance histograms.
figures/mc_ipr_histograms.png	Figure 4: IPR histograms.
figures/mc_xi_histograms.png	Figure 5: localisation-length histograms.
figures/mc_voltage_envelope.png	Figure 6: voltage-profile overlay.

Appendix B. Software environment

Component	Specification
Host OS	macOS (Darwin 25.3.0)
Simulator	LTspice (Analog Devices) at /Applications/LTspice.app/Contents/MacOS/LTspice
Python	3.13.x in virtual environment ~/ssh_robustness_env
Key Python packages	PyLTSpice, numpy, scipy, pandas, matplotlib, reportlab
AC sweep directive	.ac dec 1000 80k 400k
Random number generator	numpy.random.default_rng (PCG64)
Random seed scheme	seed = 10,000 + 10·σ_{id} + r, r ∈ [0, M); deterministic.